❑ How do you see the future potential for professionals in this area? What are the main drivers that determine how this sector/profession grows? Are you expecting some major shifts/changes, challenges and developments?
Currently, the industry has been hit by the recession quite badly. In my 25 years experience I have seen 8 or 9 downturns but this has been the worst. Though as things turn around, the industry will see growth once again.
There are a lot of technological bottlenecks that currently exist — bottlenecks that can be addressed through new and innovative chip designs. The two fields of research and application that are specially going to lead the boom are motion-sensing (MEMS) technologies and green technologies. While motion sensing will most likely produce better bio-medical equipments, efforts are on to produce chips that will help generate electricity more efficiently from renewable energy sources.

❑ Are there things one could do to progress and grow faster in this profession?
That's a huge problem amongst youngsters these days. All they want to do is to grow fast. Until recently when the economy was doing well, the attrition rate was extremely high. So many people would keep jumping from one semiconductor firm to another. Money was a huge draw. Right now, every one is lying low.

In our company we have two growth ladders — one is called the technology ladder, the other the managerial ladder. Almost everyone wants to jump to the managerial ladder as they want to grow to take control. But they forget that the future in this industry lies in understanding a particular technology and then becoming a domain expert. With time, as the industry specialises, the demand for domain experts is going to increase. Hence the pay for such people would increase too. So engineers need not abandon their love for technology to earn more. One should focus in growing in the direction of expertise.

❑ But if it is all about specialisation, then how easy is it to change careers after some time? Are there natural cross-roads? What are some usual transitions people make/can make?
The only natural cross road I can think of is that of switching from the chip design industry to the EDA tool industry. Apart from that, it is an industry for those that are interested in applied research. People who are patient and diligent and want constantly to be on the cutting edge of technology. As I said earlier, they can become domain experts after spending a long time, and the industry needs such people.

❑ Are there some institutions/ universities/ courses where you have generally found better candidates for your company?
Well, we do go to the IITs and the NITs to pick up the electronic and communication engineers. Apart from that, there are also some good private engineering colleges that give out good students.

Risks
One of the major risks in this career is the fact that it can be a little too specialised. If one wants to switch to some other industry, this can be difficult. In the first few years one can get to know if one is suited for the industry or

Major employers in India
Texas Instruments
Intel
ST Microelectronics
Samsung India
Sony Ericsson
Nokia
Cadence
Synopsis
Mentor Graphics